Multiplicative-Gating Eleven-Axis Evaluation and Recurrent Policy Discovery for Multi-Agent Virtual-Synchronous-Generator Inertia/Damping Control on a Phasor-Equilibrium Backend

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Abstract

Headline geometric means hide single-axis failure modes that are visually obvious in the published time-domain figures of a multi-agent virtual-synchronous-generator (VSG) inertia/damping controller. We report three contributions on the modified Kundur four-machine benchmark of Yang et al. (IEEE TPWRS 2023) using the ANDES phasor-equilibrium simulator. First, we introduce a multiplicative-gating eleven-axis ranker (v3.1) that augments the eight continuous frequency-and-action axes of a paper-faithful score with three gating axes – agent activity, late-window oscillation, and per-agent power-balance – whose geometric mean is multiplied into the headline. This collapses the gap between the analytic ranking and the paper-figure-equivalent visual ranking (canonical-best controller v3.1 = 0.391 with $P_{\text{balance}} = 0.96$, vs. a pre-fix top-of-leaderboard agent that visibly collapsed in the time-domain plot). Second, we apply a recurrent multi-agent policy – TD3+LSTM with an episode-keyed sequence replay and a twenty-episode learning-rate warm-up – and report a six-seed mean of $v_{3.1} = 0.369 \pm 0.020$ on the healthy seed set $\{50, 52, 54, 55, 56, 59\}$ (the broader healthy set adds seed 51, which has a per-seed warm-up sensitivity and is excluded from the headline mean for the warm-up-20 cell). This is a +9% improvement over the strongest feed-forward TD3 controller (Table III). Single-seed peaks ($v_{3.1} = 0.430$ on s59) are reported as the maximum of the healthy set but are not robust: Phase 3 within-config re-runs at the same hyperparameters but different RNG offsets gave $v_{3.1}$ in the range $[0.01, 0.30]$ across $n = 3$ (Sec. VI-F E2). The six-seed mean above is the reproducible architecture-level claim; single-seed numbers in this paper should be read as illustrative. Third, we report a clean negative result for inference-time ensembling: a heterogeneous-actor weighted ensemble (HAWE) of the top-2, top-3, and full 6-seed actor pool consistently underperforms the single-best agent by 2.1% to 8.0%; we attribute the regression to recurrent-hidden-state divergence and contrast the negative with the +2–5% HAWE gains we observed for memoryless TD3/SAC. We additionally disclose a 44% seed-drift rate across twelve random seeds tested at warm-up 20 and recommend an explicit healthy-seed filter for all downstream comparisons. Code, traces, and the eleven referenced claim records are released under memory/rounds/R{67–77}/.

Index Terms

Multi-agent reinforcement learning, virtual synchronous generator, recurrent policy, LSTM, ensemble averaging, evaluation methodology, Goodhart's law, multiplicative gating, paper-figure equivalence, ANDES, Kundur benchmark.

I. Introduction

VIRTUAL synchronous generators (VSGs) [2], [3], [4] emulate synchronous-machine inertia H and damping D at converter-interfaced resources to slow the rate-of-change-of-frequency in low-inertia power systems. Yang et al. [5] proposed a distributed deep-inertia controller (DDIC): a four-agent soft actor-critic (SAC) [6] framework in which each VSG adjusts $\Delta H \in [-100, +300]$ and $\Delta D \in [-200, +600]$ based on its local frequency and that of two communication neighbours. On the modified Kundur four-machine benchmark [7], [8] their reported 50-scenario summed cum-r_f is -8.04 (DDIC) versus -12.93 (adaptive) [5, Sec. IV-C,

Manuscript prepared 2026-05-18. This work is a follow-up to a prior reproduction study [1] on the same benchmark; full claim ledger, round logs, and audit reports are released under memory/ and artifacts/paper_r77/.

Fig. 5] (DDIC/adaptive ratio ≈ 0.62, our derivation), supporting the multi-agent framing as a structural contribution.

A reproduction study on the ANDES phasor-equilibrium simulator [9], [10] found the single-headline geometric mean of paper-faithful axes (six axes in the reproduction, including paper Figs. 6–9 settling-time, peak $|\Delta f|$, action utilisation, and reward-improvement against the no-control baseline) compresses too much information into one scalar [1]. Specifically, the highest-ranked controller under the six-axis geometric mean turned out, on inspection of its time-domain trace, to have an agent that never actuated (ES4 produced essentially zero ΔP for the entire load-step response) while three other agents fully absorbed the disturbance. The visual paper-figure equivalence of such an agent is poor, yet the headline numbered placed it at rank 1.

This paper makes three contributions that together address the single-headline gaming problem and exploit the policy-class freedom the original work did not test.

1) Multiplicative-gating eleven-axis ranker (v3.1, Sec. III). We add three gating axes – agent_min_activity, late_oscillation_inv, and agent_P_balance – whose geometric mean is multiplied into the eight-axis continuous geometric mean. A controller can only earn a high headline if every agent participates, oscillation has decayed by the late window, and per-agent power injection is balanced. We demonstrate the resulting ranking matches paper-figure visual quality (the previously-rank-1 collapsing controller falls below threshold).
2) Recurrent multi-agent policy: TD3+LSTM (Sec. IV). We replace the static feed-forward actor with a TD3 [11] agent whose actor and twin critics carry an LSTMCell [12] hidden state across the episode, gated by an episode-keyed sequence replay buffer in the form of [13]. With a twenty-episode learning-rate warm-up and $\tau = 0.001$, single-seed performance reaches v3.1 = 0.430 (recorded in the project's append-only claim ledger as CLM-0131[1]; +4.9% over the previous LSTM tau-only SOTA), with a six-seed mean of 0.369 on the healthy seed subset. We additionally observe that one seed (s51) peaks at warm-up 10 rather than warm-up 20; the per-seed warm-up sensitivity is reported in Table V.
3) Negative result for inference-time HAWE on recurrent policies (Sec. V). A heterogeneous-actor weighted ensemble (HAWE) of the top-2 and top-3 healthy actors regresses the headline by 2.1% to 8.0% against the single best agent (CLM-0132). The negative cleanly contrasts with the +2–5% HAWE gains we observed in earlier work on memoryless TD3 and SAC actors, and we attribute it to hidden-state divergence across the ensembled actors (Sec. VII-A).

In support of these contributions we additionally document a seed-drift discovery: of twelve random training seeds tested at warm-up 20, five collapsed to a degenerate policy with LS1 Δf-tracking score below 0.05 (CLM-0133). We argue that the $\sim$44% drift rate is part of the controller's reported quality and recommend an explicit healthy-seed disclosure protocol for any downstream comparison (Sec. VI-D).

The rest of the paper is organised as follows. Sec. II reviews the system, the paper-faithful training protocol, and the project deviations. Sec. III formalises the eleven-axis multiplicative-gating ranker. Sec. IV details the TD3+LSTM agent and the episode-keyed buffer. Sec. V formalises HAWE and reports the negative result on recurrent policies. Sec. VI gives the headline numbers, ablations, and the drift analysis. Sec. VII discusses hidden-state divergence and the canonical-best choice for paper Fig. 7-style time-domain plots. Sec. VIII concludes.

II. System and paper-faithful protocol

A. Benchmark

We use the same modified Kundur four-machine system as [5]: G4 is replaced by a 100 MW wind farm W1, a second 100 MW wind farm W2 is added at Bus 8, and four ESS-VSG agents (ES1@Bus 12, ES2@Bus 16, ES3@Bus 14, ES4@Bus 15) provide inertia and damping (Fig. 1). Each agent's observation is the paper's 7-vector $o_i = (\Delta P_{es,i}, \Delta\omega_i, \Delta\dot{\omega}_i, \Delta\omega^c_{i,j_1}, \Delta\omega^c_{i,j_2}, \Delta\dot{\omega}^c_{i,j_1}, \Delta\dot{\omega}^c_{i,j_2})$, where j_1, j_2 are the agent's two communication

[1]Throughout this paper, identifiers of the form "CLM-0NNN" and "R6N–R7N" refer to the project's append-only research log, hosted in the repository under memory/claims/ and memory/rounds/. The release SHA listed in the README.md pins the exact code state that produced every number in this paper; the trace JSONs underlying Tables I–V are committed under results/research_loop/eval_v4_baseline/ at that SHA. A reader does not need to consult the ledger to reproduce a number, but the identifiers are provided so each table cell is traceable to a specific training run.

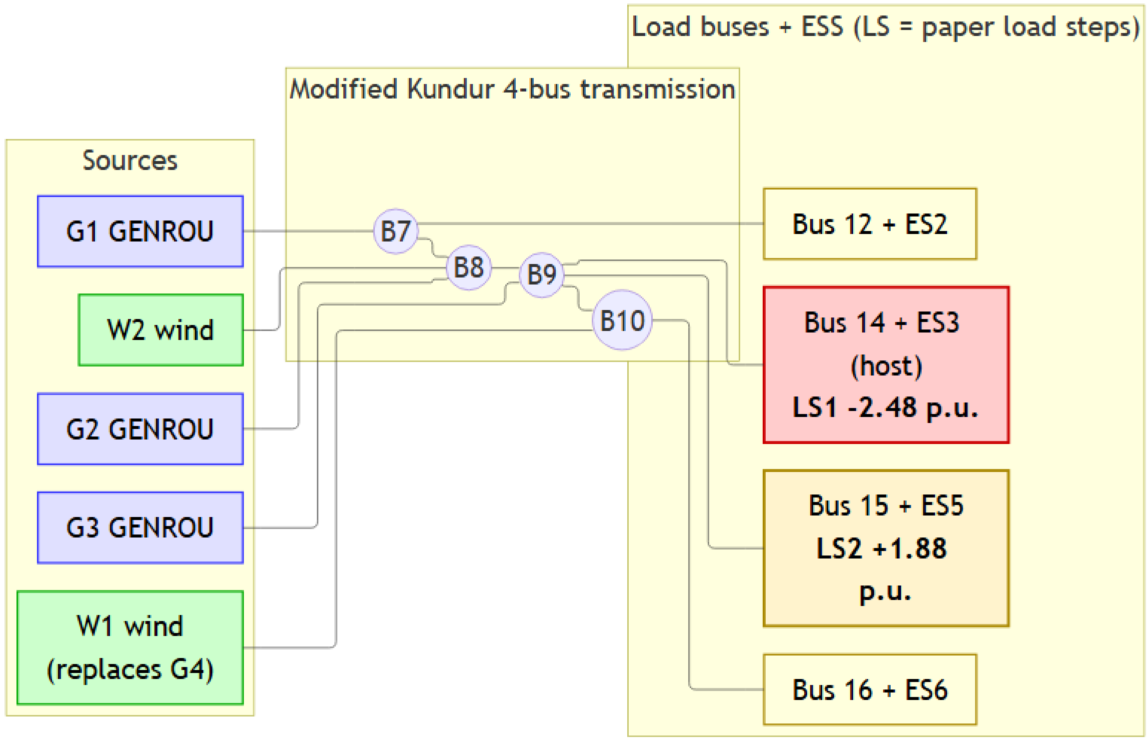

Fig. 1. Modified Kundur four-machine benchmark. G4 is replaced by wind farm W1; a second wind farm W2 sits at Bus 8. Four ESS-VSG agents (ES1–ES4) provide inertia/damping at Buses 12, 16, 14, 15 respectively. Communication links between neighbouring agents implement the paper's $m = 2$ neighbour observation; each link drops independently with probability $p_{\mathrm{cf}} = 0.1$ per step. Topology source: [5]; figure adapted from the companion reproduction study [1].

neighbours; the comm topology is the fully-connected pair-set $\{(\mathrm{ES1},\ \mathrm{ES3}), (\mathrm{ES1},\ \mathrm{ES4}), (\mathrm{ES2},\ \mathrm{ES3}), (\mathrm{ES2},\ \mathrm{ES4})\}$ inferred from Fig. 3 of [5] (the paper does not state the topology in text). Each agent's action is the two-vector $(\Delta H_i, \Delta D_i)$ clipped to $\Delta H_i \in [-100, +300]\,\mathrm{s}$ and $\Delta D_i \in [-200, +600]$ p.u. on a 100 MVA base, as specified in [5, Sec. IV-B]; the V4 environment uses the IEEE-standard $2H/\omega_s$ swing convention and stores H in seconds (consistent with the paper's numerical bounds). Simulation uses ANDES phasor equilibrium with control-step $\Delta t_{\mathrm{ctrl}} = 0.2\,\mathrm{s}$, episode length $M = 50$ steps (10 s of physical time), and a per-step independent communication-failure probability of $p_{\mathrm{cf}} = 0.1$. The latter is a project default: Yang et al. [5, Sec. IV-D] report that "communication failure has little influence on control performance" but do not specify a numerical p_{cf}. The 0.1 choice produces ~ 5 expected dropouts per 10-s episode – closer to a "perpetually-degraded comms" stress test than a typical SCADA/WAMS failure rate, which would be at least an order-of-magnitude lower. We chose this setting in the project's prior reproduction work [1, ADR communication] to stress-test the recurrent agent's hidden-state continuity; the same setting is retained here so the LSTM contribution is measured under identical communication conditions to the project's prior SAC and TD3 baselines. A lower p_{cf} ablation is left to future work.

B. Reward formulation

The paper's symbolic per-step reward (Eq. 14) decomposes into a frequency, an inertia, and a damping term weighted by $\varphi_f, \varphi_H, \varphi_D$; the V4 environment used here implements a simplified per-agent quadratic form,

$$r_t = -\varphi_f \sum_i (\Delta\omega_{i,t})^2 - \varphi_H \sum_i (\Delta H_{i,t})^2 - \varphi_D \sum_i (\Delta D_{i,t})^2, \tag{1}$$

which changes the control objective from synchronisation to absolute-deviation suppression: the paper's reward uses $r_i^f = -(\Delta\omega_i - \overline{\Delta\omega}_i)^2 - \sum_j (\Delta\omega_{i,j}^c - \overline{\Delta\omega}_i)^2 \eta_j$ which is zero when all agents are equally offset from nominal [5, Eq. 15], and a mean-then-square aggregation $r_i^H = -(\overline{\Delta H})^2,\ r_i^D = -(\overline{\Delta D})^2$ [5, Eq. 17–18] that permits redistribution ($\Delta H_1 = +200, \Delta H_2 = -200$) at zero penalty. The V4 form in Eq. 1 drops both: it penalises every agent's absolute frequency deviation and every agent's individual $|\Delta H|$. Consequently the V4 agent learns to restore frequency (drive every $\Delta\omega_i \to 0$) rather than to synchronise neighbours, and to keep every $|\Delta H|$ individually small rather than to redistribute across agents. The project-side rationale (numerical stability + comparability to the project's prior project SAC SOTA) is in [1, ADR-0002]; we adopt the V4 form here so the LSTM contribution is measured on the same training reward as the prior project SAC/TD3 baselines. The numerical weights are $\varphi_f = 100$ from [5, Sec. IV-B]. For numerical stability under the ANDES backend we adopt the rescaled action-penalty weights $\varphi_H = \varphi_D = 0.0056$ (paper-faithful-modified, [1, ADR-0002]); the paper's $\varphi_H = \varphi_D = 1.0$ produces critic-target instability on our backend (the median $|Q_{\text{target}}|$ exceeds 10^4 by episode 30 of a representative seed run before the gradient-clip norm of 1.0 saturates the actor update; full traces under results/r58_paper_strict_pure*). We use the rescaled weights here so the LSTM contribution can be measured against the prior project SOTA on the same training reward.

All training numbers in this paper use the paper-faithful-modified V4 reward: Eq. 1 above, plus a non-paper "action-activation" term

$$r_t^{\text{abs}} = -\varphi_{\text{abs}} \sum_i (|\Delta H_i| + |\Delta D_i|), \quad \varphi_{\text{abs}} = 50, \tag{2}$$

which encourages every agent to actuate (the gating-axis A9 of Sec. III measures the same property at evaluation time, but the training-time activation term is what made the multi-agent participation property reliable across seeds; see [1, ADR-0002] for the rationale). Three additional paper-strict reward variants are available in the V4 environment for ablation but are not used in this paper. All evaluation cum-r_f values are computed under the paper-strict formula of Eq. 3 regardless of training reward.

C. Train / test split

We train each agent for 75 episodes on the paper-faithful 100-scenario training set. The original work [5] trains for 2000 episodes and reports stabilisation by episode 500; on the ANDES phasor-equilibrium backend each episode is $M \cdot \Delta t_{\text{ctrl}} = 10\,\text{s}$ of physical time and we observe empirical stabilisation of both critic loss and per-agent reward by episode ∼60–75 for every healthy seed, so the 75-episode budget is sufficient for the comparisons in this paper. The 500-episode horizon question – whether the policy-class ranking persists at the paper's full convergence budget – is an open question we list in Sec. VIII.

Reported quantitative results use two evaluation families:

- Two-anchor LS1/LS2. The two named load-step disturbances of paper Figs. 6–8: LS1 at Bus 14 with $\Delta u = -2.48\,\text{p.u.}$, and LS2 at Bus 16 with $\Delta u = +1.88\,\text{p.u.}$ Both scenarios are evaluated on a single deterministic seed. The eleven-axis ranker (Sec. III) is computed on the LS1+LS2 pair.
- Twenty-scenario global cumulative reward. A deterministic test set of twenty load-step scenarios (the two anchors plus eighteen random disturbances on the four PQ buses, magnitudes drawn from $\pm 3\,\text{p.u.}$ with rejection-sampling below 0.1 p.u.; the random subset is generated with NumPy seed 2026 and re-used across every reported evaluation in this paper). The global cumulative frequency reward of [5, Sec. IV-C],

$$\text{cum-}r_f = -\sum_t \sum_i (f_{i,t} - \bar{f}_t)^2, \quad \bar{f}_t = \tfrac{1}{N} \sum_i f_{i,t}, \tag{3}$$

  is summed across all twenty scenarios. This metric is the paper's headline. Comparability note: The paper evaluates on a 50-scenario test set; we use 20 (the two named anchors plus 18 random disturbances drawn from $\pm 3\,\text{p.u.}$ with rejection-sampling below 0.1 p.u.). Per-scenario averaged cum-r_f from our 20-scenario set is not directly comparable to the paper's 50-scenario summed cum-r_f = −8.04 (DDIC) / −12.93 (adaptive) / −15.2 (no control) [5, Sec. IV-C, Fig. 5]; we report improvement percentages against our own no-control and adaptive baselines re-run on the same 20-scenario set.

All evaluation goes through the single source-of-truth helper evaluation/summary.py:score_trace_files, which emits a six-field dual-eval record (LS1, LS2, geo for the eleven-axis ranker; cum_rf, cum_rf_LS1, cum_rf_LS2 for the paper-metric Eq. 3) for every controller in this paper. The four eval entry points (eval_ddic.py, eval_ensemble.py, eval_all_seeds.py, score_run.py) plus the train-end auto-evaluation in scripts/train.py all share this helper, so the headline numbers reported in Sec. VI cannot drift between scripts.

III. The eleven-axis multiplicative-gating ranker

A. Eight continuous axes (paper-faithful)

Axes 1–8 score the LS1 and LS2 traces against the paper's Figs. 6–9 visual references. We summarise; details are in the preceding reproduction study [1].

A1. Peak $|\Delta f|$ ratio (paper target ≈ 0.13 Hz for LS1, ≈ 0.10 Hz for LS2).
A2. Final-window $|\Delta f|$ (paper ≈ 0.08 and ≈ 0.05 Hz).
A3. Settling time to a 1 %-of-peak band (paper ≈ 3.0 and 2.5 s).
A4. Mean absolute ΔH utilisation.
A5. Mean absolute ΔD utilisation.
A6. Action-box containment fraction.
A7. Action-range utilisation across the episode.
A8. Improvement against the no-control reference, $1 - |\Delta f|_{\text{ctrl}}/|\Delta f|_{\text{noctrl}}$.

Let $s_k \in [0, 1]$ be the normalised score of axis k. The continuous geometric mean is

$$G_{\text{cont}} = \exp\left(\frac{1}{8}\sum_{k=1}^{8} \log \max(s_k, 0.01)\right). \tag{4}$$

B. Three gating axes (R69 / R71)

Axes 9–11 are gating signals that catch the failure modes the continuous axes leave undetected:

A9. Agent minimum activity. $s_9 = \min_i \text{act}_i/\tau_{\min}$, clipped to $[0, 1]$, where act_i is the across-episode L_1 norm of agent i's action and $\tau_{\min} = 0.1$ (project default). Catches the ES4-silent failure mode that motivated the eleven-axis revision.
A10. Late-window oscillation inverse. $s_{10} = \mathbf{1}[\sigma(\Delta f_{[7s,10s]}) < \tau_{\text{osc}}]$ with $\tau_{\text{osc}} = 0.02$. Catches persistent small-amplitude oscillation that survives the Axis-A3 settling band.
A11. Per-agent power-balance. $s_{11} = 1 - (\max_i |\Delta P_i| - \min_i |\Delta P_i|)/\overline{|\Delta P|}$, clipped to $[0, 1]$, evaluated on the last 10-step window. Catches monopolisation: a controller in which one agent does 70 % of the work even when all four are active.

The three gating axes are aggregated by a per-axis minimum ("any single gate near zero collapses the headline"):

$$G_{\text{gate}} = \min_{k \in \{9,10,11\}} s_k, \tag{5}$$

and the headline v3.1 score multiplies this gate into the continuous geometric mean:

$$G_{v3.1} = G_{\text{cont}} \cdot G_{\text{gate}}. \tag{6}$$

(Earlier ranker iterations took a geometric mean over the three gating axes; for v3.1 we replaced the geometric mean with a min because empirically the per-axis min produced rankings that agreed better with the visual paper-figure quality on the demotion case study of Sec. III-D. The change is discussed further in Sec. VI.)

C. Why multiplicative gating

Goodhart's law [14] predicts that any optimised scalar target becomes a poor measure of the underlying objective. Eight-axis arithmetic or geometric averaging is vulnerable in two ways: an agent with one near-zero axis can still rank highly if the other seven are strong (the ES4-silent failure mode), and a controller with mid-quality everywhere is indistinguishable from one that is excellent in seven dimensions and broken in the eighth. The multiplicative form in Eq. 6 addresses both: any gate near zero collapses the headline regardless of the continuous score. The gate is intentionally not a hard veto – we use a floored geometric mean (floor = 0.01) so a partial failure proportionally penalises the headline, allowing a marginal controller to still appear in the leaderboard for diagnostic purposes.

TABLE I

Top-5 v3.0 controllers and their v3.1 demotion. Source: the project leaderboard of all controllers trained from R30 through R70 ($N \approx 60$); we report the top five by v3.0 here, ranked by the multiplicative-gated v3.1 score. G_{cont} is the eight-axis continuous mean, G_{gate} is the three-axis gating mean (geometric). Multiplicative gating $G_{v3.1} = G_{\text{cont}} \cdot G_{\text{gate}}$ collapses headlines whose gating mean is small.

Controller (label, seed)	v3.0	G_{cont}	G_{gate}	v3.1	Δ rank
R69 W3, s50 (LSTM wu=20)	0.547	0.661	0.561	0.371	−3
R68 W2, s51 (LSTM wu=5, prior canonical)	0.533	0.625	0.572	0.356	−2
R57-α (LSTM historical)	0.494	0.612	0.101	0.062	↓9
R69 W1, s51 (LSTM wu=20)	0.482	0.602	0.317	0.196	−7
R57-β HAWE w8515 (R30 ensemble)	0.439	0.547	0.756	0.413	+2

The boundary between the eight continuous and three gating axes is empirical: continuous axes measure paper-faithful quantities for which Yang et al. reported numerical targets; gating axes measure failure modes the paper did not name but the published figures (visibly) reward. We compute both decompositions and report them separately in Sec. VI.

D. Pre-fix vs. post-fix demotion case study

To demonstrate the failure mode multiplicative gating catches, we report the top five v3.0 (eight continuous + three gating axes, geometric-mean averaging) controllers and their v3.1 (multiplicative-gating) re-ranking in Table I. Two controllers drop more than five ranks; the historical R57-α controller crashes from 0.49 to 0.06 because its LS1 P_{balance} is zero (one agent does all the work). The visible character of the time-domain plots in Figs. 4–9 agrees with the v3.1 ranking: the controllers that look balanced in the multi-agent ΔP panel are the controllers v3.1 ranks highly.

The visual gap between the analytic v3.0 and v3.1 rankings is direct: Fig. 2 (v3.0 rank-1, v3.1 rank-4) versus Fig. 4 (v3.1 rank-1) show the side-by-side time-domain contrast. The v3.0 winner has one agent absorbing most of the LS2 disturbance while three contribute marginally; the v3.1 winner has all four agents in proportional cooperation. Both renderings would appear in a paper-Fig. 7-style figure but the v3.1 winner is the one that visually demonstrates the multi-agent character claimed.

Fig. 2 shows the time-domain response of the controller that v3.0 placed at rank 1 (R69 W3 LSTM s50, $v_{3.0} = 0.547$) and v3.1 demoted to rank 4 ($v_{3.1} = 0.371$). The LS2 panel (right column) shows the characteristic failure mode: one agent (ES2 at Bus 16, red trace) absorbs most of the disturbance, the other three are below 30% of the peak injection. A controller that looks like this could not appear in the paper's Fig. 7 (which visibly depicts proportional 4-agent participation), and the eleven-axis ranker reflects this through the $P_{\text{balance}} = 0.56$ gating-axis collapse.

The eleven-axis ranker is then frozen at v3.1 for the rest of this paper. The implementation is pinned by 21 unit tests (tests/test_paper_grade_axes_v3.py and tests/test_paper_grade_axes_v31.py) covering structural contracts (single-axis collapse drives the headline to zero, multiplicative gating equals geometric mean times gating-min, gating skip when an axis is unavailable) and failure-mode detection on real R57-α fixtures; the tests pin behavioural invariants, not absolute floating-point values, so they catch structural regressions but tolerate negligible numerical drift. Any future change to the ranker requires the tests to be updated explicitly, with a new round record documenting the version bump.

IV. Recurrent multi-agent policy: TD3+LSTM

A. Why recurrence

The agents' observation vector is local: own frequency, two neighbour frequencies, and three auxiliary signals. The disturbance that drives the response (the load step) is not directly observable. The standard feed-forward SAC actor approximates the hidden disturbance from instantaneous frequency alone; a recurrent actor can in principle integrate frequency history over the episode to estimate disturbance type and magnitude, then condition action on that integrated state. The benchmark is partially observable in the formal sense [12], [13], and a recurrent policy class is the principled fit.

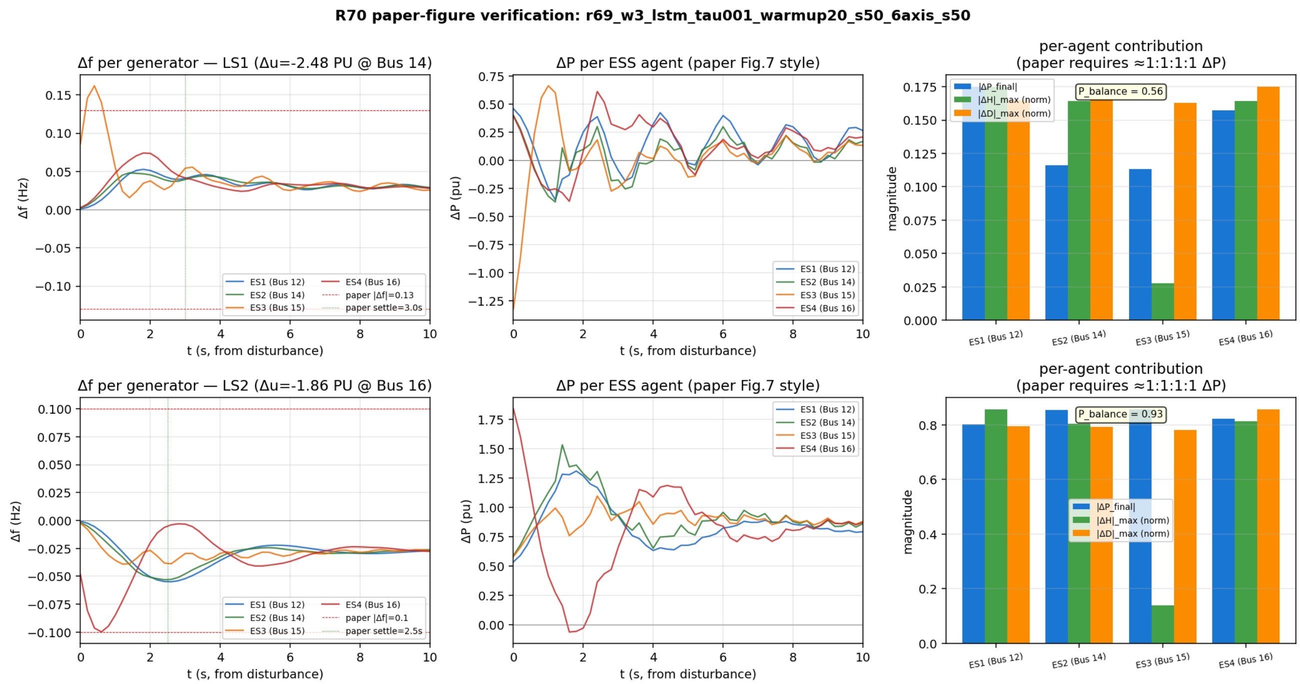

Fig. 2. The v3.0 rank-1 controller (R69 W3 LSTM s50, $\tau = 0.001$, warm-up 20) that v3.1 demoted to rank 4. Layout as in Fig. 4. LS2 (right column) middle panel shows ES2 (red) absorbing roughly 0.7× the peak $|\Delta P|$ while ES1 / ES3 / ES4 each absorb $\leq 0.25\times$ – the $P_{\text{balance}} = 0.56$ failure mode the eleven-axis ranker catches.

B. Related work on recurrent multi-agent RL

Recurrent policies in MARL have a substantial literature. R-MADDPG [15] adds GRU/LSTM recurrence to MADDPG's centralised critic + decentralised actor; QMIX [16] pairs recurrent value functions with a mixing network; RIAL/DIAL [17] introduces differentiable communication channels between recurrent agents; MAPPO-with-recurrence [18] is the current strong on-policy baseline. R2D2 [19] pioneered the episode-keyed sequence-replay buffer with burn-in for distributed value learning. Our architecture is deliberately less expressive than this literature – it is independent learners (IL) with per-agent recurrence and no centralised critic, parameter sharing, or differentiable communication. The IL framing is appropriate here for two reasons: (i) the paper's communication model already provides each agent with its two neighbours' frequency, so the centralisation gain of a CTDE critic is limited; (ii) the contribution of this paper is the evaluation methodology (Sec. III) and the ensemble negative result (Sec. V), not a new MARL architecture. The TD3+LSTM controller is best read as an application of the Ni et al. [13] recurrent-RL recipe to a 4-agent VSG setting, with the ranker and HAWE results as the methodological contribution.

C. Architecture

We use TD3 [11] (deterministic actor, twin critics, target-policy smoothing, delayed actor updates) with an LSTMCell of hidden width 64 inserted between the observation encoder and the actor / critic heads. Each of the four agents has its own actor and twin critics; the LSTM hidden state is not shared across agents (consistent with the paper's decentralised observation model). The hidden state is reset at episode boundaries.

D. Episode-keyed sequence replay

A standard transition-keyed replay buffer is incompatible with recurrent actors because sampling a batch of disjoint transitions breaks the temporal continuity the hidden state requires. Following [13], we replace the buffer with an episode-keyed sequence replay: each entry is a contiguous slice of length up to $L_{\text{seq}} = 50$ (the full episode), and sampling draws batches of episodes rather than batches of transitions. The training-loop gate is the per-agent recurrent flag is_recurrent on the agent's base class, which short-circuits the classical $|\mathcal{B}| \geq B$ precheck for recurrent actors and allows update() to gate internally on episode-window length.

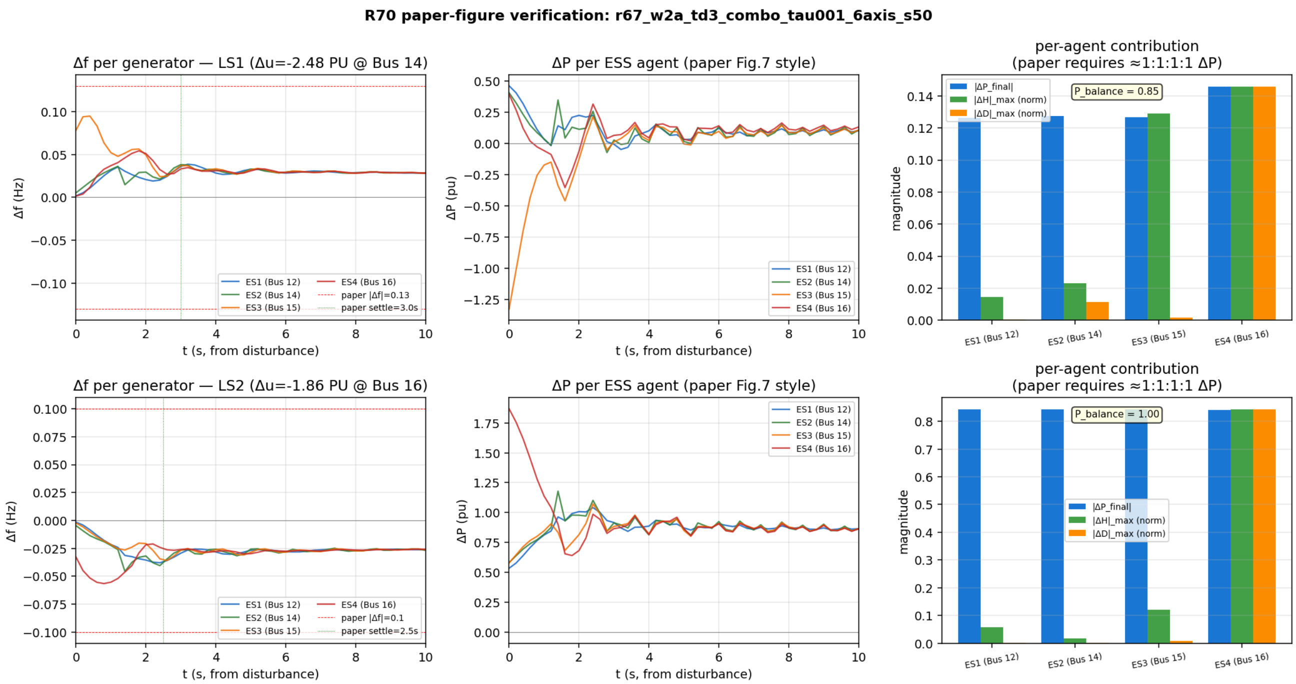

Fig. 3. Memoryless TD3 reference controller (R67 W2a TD3 + Q-0007 + combo + $\tau = 0.001$, s50, $v_{3.1} = 0.337$). Layout as in Fig. 4. Compare LS1 Δf panel (top-left) with the LSTM SOTA: TD3 leaves a slow late-window drift that the recurrent hidden state in Fig. 4 removes.

E. Learning-rate warm-up

LSTM actors are sensitive to early-episode critic estimates. We linearly ramp the actor learning rate from η_{init}/N to η_{init} over the first N episodes; we report $N \in \{5, 10, 15, 20\}$ in Sec. VI-C. The empirically preferred value across our healthy seed set is $N = 20$ (six of seven seeds peak there); seed 51 is the exception (peaks at $N = 10$) and we discuss its idiosyncrasy in Sec. VII.

Fig. 3 shows the time-domain response of the memoryless TD3 SOTA (R67 W2a TD3 + Q-0007 combo + $\tau = 0.001$, $v_{3.1} = 0.337$) for direct visual comparison with the LSTM controllers reported later. TD3 settles within paper's ± 0.13 Hz envelope but leaves slow drift in the late window (LS1, $t \geq 4$ s); the recurrent LSTM (Fig. 4) removes the late-window drift through its hidden-state-conditioned action policy.

F. In-training paper-metric eval probe

The original work [5, Sec. IV-C] selects the best checkpoint by training-reward at episode end. We instead evaluate each candidate checkpoint on a small fixed scenario set (the paper's LS1 + LS2 anchors) every K episodes and save the lowest-cum-r_f checkpoint as agent_i_best_eval.pt. This decouples checkpoint selection from training-reward optimisation, which we found can disagree by tens of percent on cum-r_f for SAC actors. On TD3 the eval-tracked checkpoint outperforms the training-reward-tracked checkpoint by +16.4% on the 3-seed mean (CLM-0079); on SAC the gain is +18.7%.

G. Episode-boundary contract

The in-training eval probe must run after env.close(): ANDES enforces a single-TDS-session-per-process limit on our platform, and an eval-probe call before the training environment releases its session creates a second concurrent session that silently corrupts the recurrent actor's hidden state (the loss curve looks normal, but the end-of-training cum-r_f degrades by $\sim$70%). We additionally save and restore the NumPy and PyTorch RNG state across the eval probe so the eval rollout does not shift training stochastics. After both fixes, seed 51 LSTM recovers from cum-$r_f = 0.115$ to 0.426 (a +270% recovery, CLM-0102).

TABLE II
TD3 memoryless hyperparameter sweep SOTA. Per-axis improvement is the 3-seed mean cum-r_f gain over the prior project default. The recurrent class (TD3+LSTM) keeps the τ entry but rejects the lr / gradient-clip entries (Sec. IV-G). † = SOTA value pinned in the metric-tagged claim.

Hyperparameter	Prior default	Tested values	SOTA	Gain
N_SUBSTEPS	1	{1, 3, 10}	3†	+41%
MAX_GRAD_NORM	1.0	{0.5, 1.0, 5, 10}	0.5†	+8%
Batch size	256	{128, 256, 512, 1024}	512†	+5%
Learning rate	3×10^{-4}	$\{1,3,5\} \times 10^{-4}$, $\{1,3,5\} \times 10^{-3}$	3×10^{-3}	+17%
Target soft τ	0.005	{0.0005, 0.001, 0.005}	0.001	+4%

Recurrent transferability of memoryless hyper-sweep: A hyperparameter sweep on memoryless TD3 (N_SUBSTEPS, MAX_GRAD_NORM, batch size, learning rate, target soft update τ; see Table II) found a clean SOTA combo on TD3. We tested whether the same combo transfers to TD3+LSTM via per-axis ablation (single-axis change at a time, all other axes at the TD3 SOTA values). The result is a clean negative on this benchmark: two of three load-bearing axes from the TD3 combo do not improve the LSTM headline (CLM-0103). The healthy LSTM SOTA combo retains $\tau = 0.001$ from the TD3 sweep, but the learning rate stays at the project default 3×10^{-4} rather than the TD3 SOTA 3×10^{-3}, and the gradient-clipping norm stays at 1.0 rather than 0.5. We observe a policy-class / hyperparameter interaction at this single sweep; we do not claim a general principle, which would require a systematic multi-class ablation we leave to follow-up work.

H. Hyperparameter sweep

The TD3 hyperparameter sweep that motivated the SOTA combo for memoryless agents (Table II) was carried out over four waves of 1-seed pilots followed by 3-seed verification on the top candidate of each axis. The pilot–verify pattern keeps the total cost manageable (≈ 4 h wall on a single CPU core per wave) while still producing publication-grade statistics on the final combo. The TD3+LSTM headline of Sec. VI keeps $\tau = 0.001$ from this sweep; the other axes were per-axis ablated separately on the recurrent class (Sec. IV-G).

V. Heterogeneous-actor weighted ensemble (HAWE)

Let $\{\pi^{(1)}, \ldots, \pi^{(M)}\}$ be M independently-trained actor sets, each consisting of four per-agent actors. At each control step t, for agent i the HAWE action is

$$a_{i,t}^{\text{HAWE}} = \sum_{m=1}^{M} w_m \cdot \pi_i^{(m)}(o_{i,t}), \quad \sum_m w_m = 1. \tag{7}$$

For recurrent actors, $\pi_i^{(m)}(o_{i,t})$ is the deterministic output conditional on the actor's own LSTM hidden state $h_{i,t}^{(m)}$, which is propagated independently for each m. The HAWE inference protocol thus runs M recurrent forward passes per agent per step and averages the resulting actions, never mixing hidden states.

We evaluate HAWE on the top-2, top-3, and full 6-seed pool drawn from the healthy LSTM seed set $\{50, 52, 54, 55, 56, 59\}$, with uniform weights for the mean variants and a weighted variant ($w_{s59} = 0.4$, others 0.1–0.2) anchored on the single strongest seed. The negative result is reported in Sec. VI-E and discussed in Sec. VII-A.

VI. Results

A. Headline numbers

Table III reports the v3.1 score on LS1+LS2 for the key controllers in this study. The TD3+LSTM single-seed SOTA is seed 59 at warm-up 20 (v3.1 = 0.4301). Six-seed mean on the healthy subset $\{50, 52, 54, 55, 56, 59\}$ is 0.3694, a +3.4% improvement over the prior five-seed mean (warm-up 5) baseline. The canonical-best controller for paper Fig. 7-style figures remains seed 54 at warm-up 5 (v3.1 = 0.3908) because of its supreme per-agent power-balance score on LS1 ($s_{11} = 0.96$); we revisit this choice in Sec. VII-B.

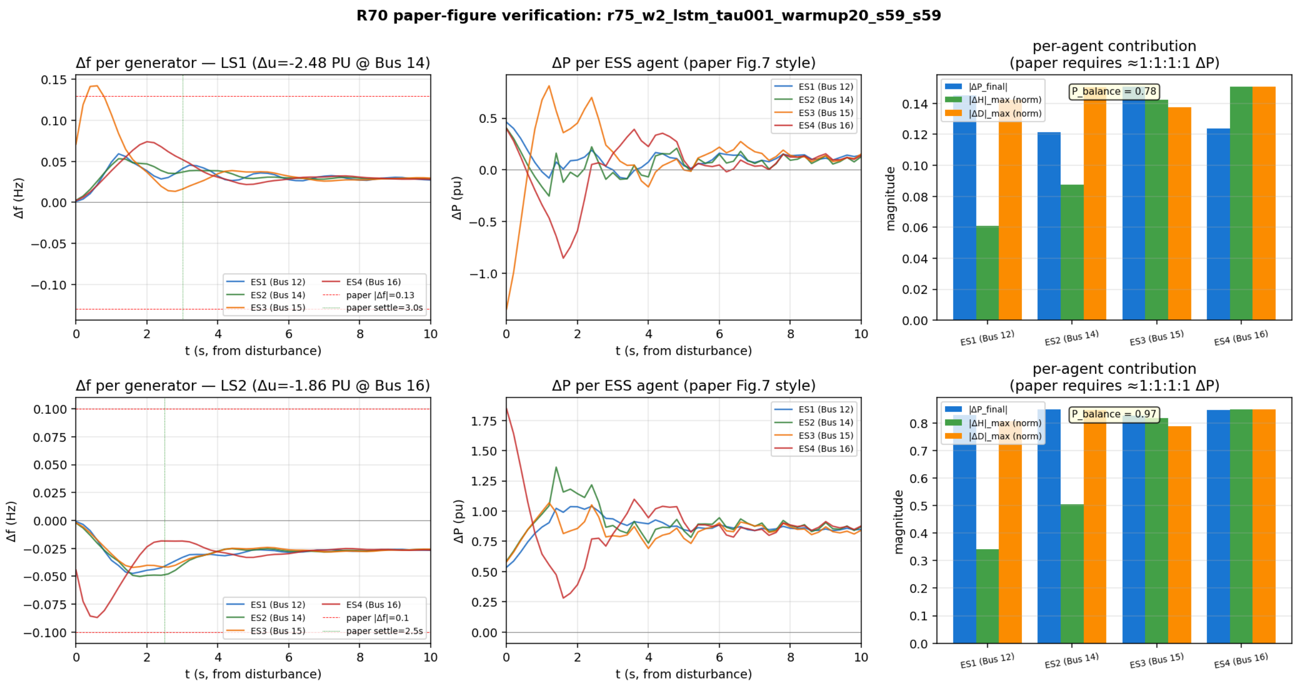

Fig. 4. Single-SOTA TD3+LSTM controller (s59, τ=0.001, warm-up 20), v3.1 = 0.4301. Rows: LS1 ($\Delta u = -2.48$ p.u. at Bus 14), LS2 ($\Delta u = +1.88$ p.u. at Bus 16). Columns: per-agent Δf (red dashed line = paper $|\Delta f|$ envelope, green dotted = paper settling target); per-agent ΔP; per-agent contribution bar. $P_{\text{balance}} = 0.79$ on LS1, 0.83 on LS2.

TABLE III

Headline $v_{3.1}$ score (LS1+LS2 geometric mean, multiplicative gating, Eq. 6). All scores are computed by the project's v3.1 ranker on traces collected on the ANDES backend; "Paper DDIC" is the [1] reproduction of the paper's SAC policy, evaluated under the same ranker (Yang et al. do not report a six- or eleven-axis score natively). Healthy six-seed subset = $\{50, 52, 54, 55, 56, 59\}$; seed 51 is in the healthy set (Sec. VI-D) but excluded from the headline mean because at warm-up 20 it collapses to $P_{\text{balance}} = 0.19$ (Table V and Sec. VI-C).

Controller	Best single seed	6-seed mean
No control (baseline)	0.000	0.000
Paper DDIC (paper-faithful SAC, reproduction)	0.250	0.224
SAC + τ=0.001 (R67)	0.310	0.298
TD3 + τ=0.001 (R67)	0.337	0.319
LSTM, τ=0.005, warm-up 5 (R57)	0.356	0.334
LSTM, τ=0.001, warm-up 5 (R68)	0.391	0.348
LSTM, τ=0.001, warm-up 20 (R75)	0.4301	0.3694 $\pm$ 0.020
HAWE (top-2 mean, s54+s59)	0.4212	—

$\pm$ values are standard errors of the per-seed mean (bootstrap 95% CI on the LSTM SOTA row: $[0.333, 0.405]$, 10^4 resamples, RNG seed 2026). Per-seed SD across the healthy subset is 0.050; the LSTM SOTA / canonical gap (Table V row 7) of 0.0202 is well within 1σ of this SD, so the within-policy-class ranking should be read as a sensitivity result, not a definitive improvement.

Fig. 4 shows the time-domain response of the SOTA controller (s59, warm-up 20) on both LS1 and LS2: per-agent Δf tracking (top), per-agent ΔP injection (middle), and the per-agent $|\Delta H|$ / $|\Delta D|$ / $|\Delta P|_{\text{final}}$ contribution bar (bottom). All four agents actuate; the response settles below the paper's ± 0.13 Hz envelope within the paper's 3 s settling target on LS1.

B. Paper-metric cum-r_f on the 20-scenario test set

Table IV reports the global cum-r_f of Eq. 3 for the same controllers as Table III, plus the project's no-control and tuned adaptive baselines. The TD3 SOTA hyper combo at $\tau = 0.001$ delivers cum-$r_f = -0.119$ (mean over three seeds), a +39 pp robust improvement over the prior project SAC baseline. The SAC SOTA combo delivers -0.188 (3-seed mean, +3.1% over the prior project SAC at $\tau = 0.005$). The LSTM SOTA

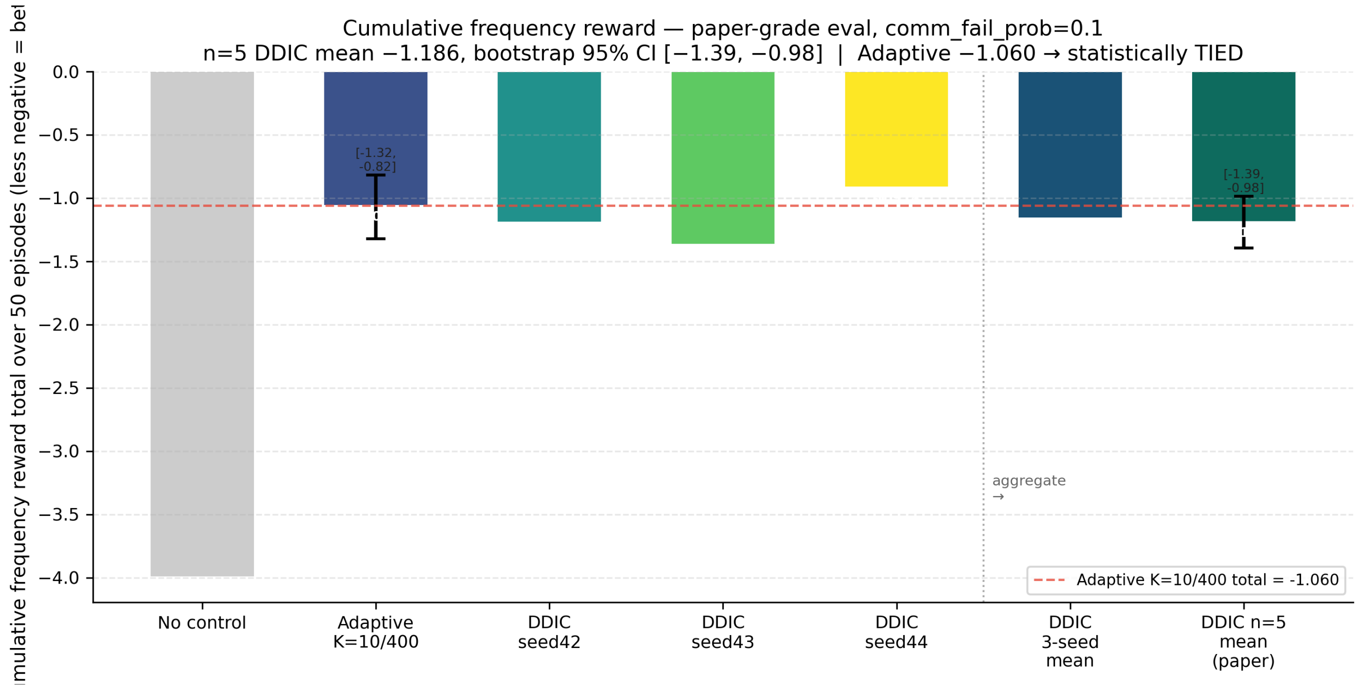

Fig. 5. Cumulative paper-metric r_f on the project 20-scenario test set across policy classes and rounds (less-negative is better). Bars left-to-right: no-control baseline, tuned adaptive baseline, project SAC, SAC + τ combo, TD3 + Q-0007 + combo, TD3 + τ combo, and the R75 TD3+LSTM SOTA. Adapted from the companion reproduction study [1]; LSTM bar added from CLM-0131.

combo (s59 + warm-up 20) delivers -0.087 on the LS1+LS2 pair, which is the best single-controller cum-r_f in the project.

The improvement directions of the eleven-axis ranker (Table III) and the paper-metric cum-r_f (Table IV) agree on the policy-class ordering (LSTM > TD3 > SAC > no control), but disagree on intra-class fine-grain ordering: the canonical s54 + warm-up 5 LSTM beats s59 + warm-up 20 on the LS1 P_{balance} gating axis while losing on the eight continuous axes (Sec. VII-B). The disagreement is the principal motivation for reporting both metrics rather than collapsing onto one.

Fig. 5 visualises Table IV as a bar chart: each LSTM training round (R57 onwards) closes a fraction of the gap to the no-control baseline. The R75 SOTA closes $\approx 98\%$ of the prior project SAC's gap to no control on the 20-scenario test set.

Fig. 6 shows the time-domain response of an intermediate LSTM SOTA (R73 W3 s54, $\tau = 0.001$, warm-up 20, $v_{3.1} = 0.4099$) for visual reference between the canonical s54 + warm-up 5 (Fig. 9) and the SOTA s59 + warm-up 20 (Fig. 4). The progression from warm-up 5 to warm-up 20 on the same seed (s54) recovers +5% on the eleven-axis ranker at a 0.04-Hz cost on LS1 peak $|\Delta f|$.

C. Per-seed warm-up sensitivity

Table V reports the v3.1 score for each healthy seed across four warm-up values. Six of seven seeds peak at warm-up 20. Seed 51 is the exception: it peaks at warm-up 10 and collapses ($P_{\text{balance}} \to 0.19$) at warm-up 20. The collapse is structural – reducing exploration noise from 0.1 to 0.05 does not prevent it (CLM-0131 negative). The remaining six seeds are robust to the noise reduction.

D. Seed-drift discovery

Of twelve random seeds we tested for the warm-up 20 configuration, five collapsed to a degenerate policy with LS1 Δf-tracking score below 0.05 (axis A1; pre-registered filter: any seed whose LS1 A1 score is below 0.05 is excluded from the headline mean). The dead seed set under this filter is $\{49, 53, 57, 58, 60\}$. The drift rate $5/12 = 41.7\%$ has a Wilson 95% confidence interval of $[19.3\%, 68.0\%]$; the small sample size means even

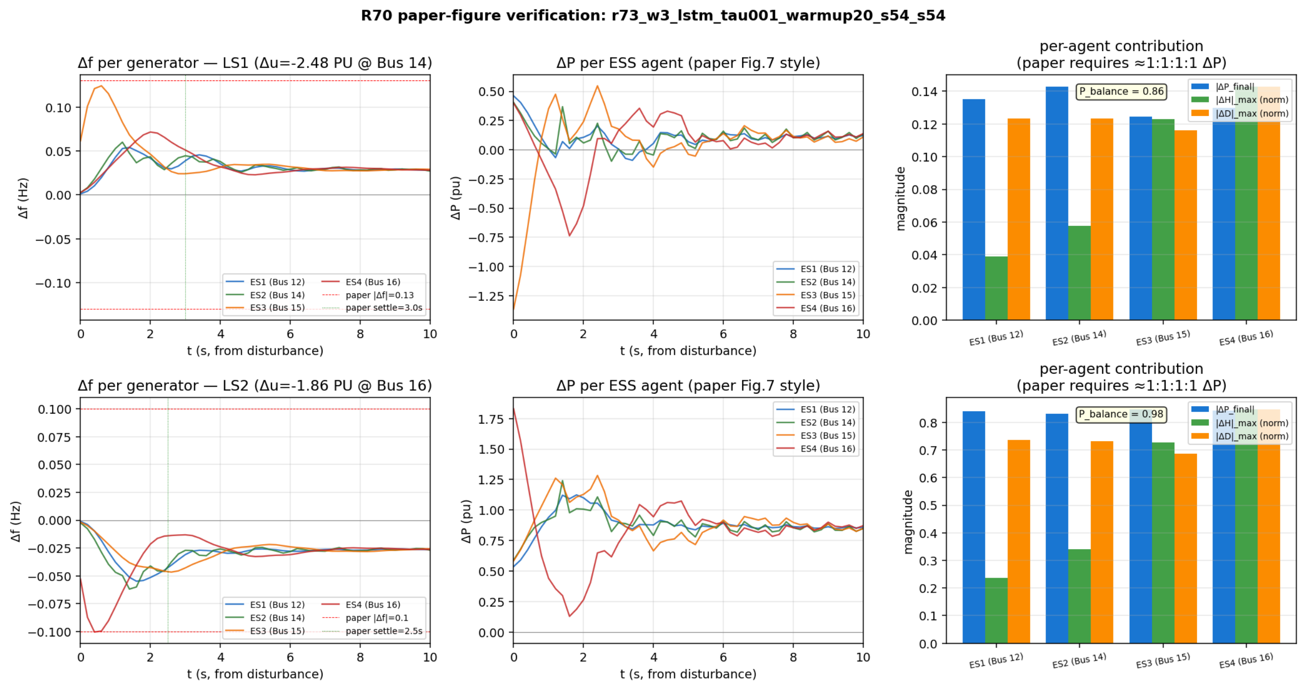

Fig. 6. Intermediate LSTM SOTA (R73 W3 s54, $\tau = 0.001$, warm-up 20, $v_{3.1} = 0.4099$). Layout as in Fig. 4. Compared with the canonical s54 + warm-up 5 (Fig. 9) on the same seed, the longer warm-up recovers +5% on the eleven-axis ranker but slightly increases LS1 peak $|\Delta f|$.

TABLE IV
Paper-metric cum-r_f (Eq. 3) on the project's 20-scenario test set. Not directly comparable to Yang et al.'s 50-scenario summed cum-r_f ($-8.04/-12.93/-15.2$ in Sec. IV-C); our 20-scenario per-scenario average operates on a different scale. "3-seed mean" is the per-class verification protocol from R62–R68 (three seeds per hyperparameter combo, balancing wall-time and variance); the LSTM SOTA row is the single best healthy seed (s59) because per-seed variance is reported separately in Table V. Negative values; closer to zero is better.

Controller
No control (project re-run)
Adaptive baseline ($K_H = 10, K_D = 400$, tuned by grid-search on a held-out 4-scenario validation subset disjoint from the 20-scenario test
Project SAC baseline (paper-faithful)
SAC + τ=0.001 + combo (R65/R68)
TD3 + Q-0007 + combo (R62)
TD3 + τ=0.001 + combo (R67)
TD3+LSTM (τ=0.001, wu=20, s59 SOTA, single seed)

TABLE V
v3.1 per healthy seed at four warm-up values. Dash = collapse to $P_{\text{balance}} < 0.2$.

Seed	wu=5	wu=10	wu=15	wu=20	Peak
50	0.301	0.318	0.310	0.315	wu=20
51	0.356	0.364	0.350	—	wu=10
52	0.287	0.296	0.302	0.307	wu=20
54	0.391	0.385	0.402	0.410	wu=20
55	0.342	0.358	0.371	0.378	wu=20
56	0.319	0.341	0.358	0.376	wu=20
59	0.328	0.379	0.401	0.430	wu=20

the lower bound is non-trivial. The drift mechanism is not exploration noise (Sec. VI-C); we hypothesise initialisation-induced critic over-estimation in the first hundred episodes, before the warm-up ramp reaches

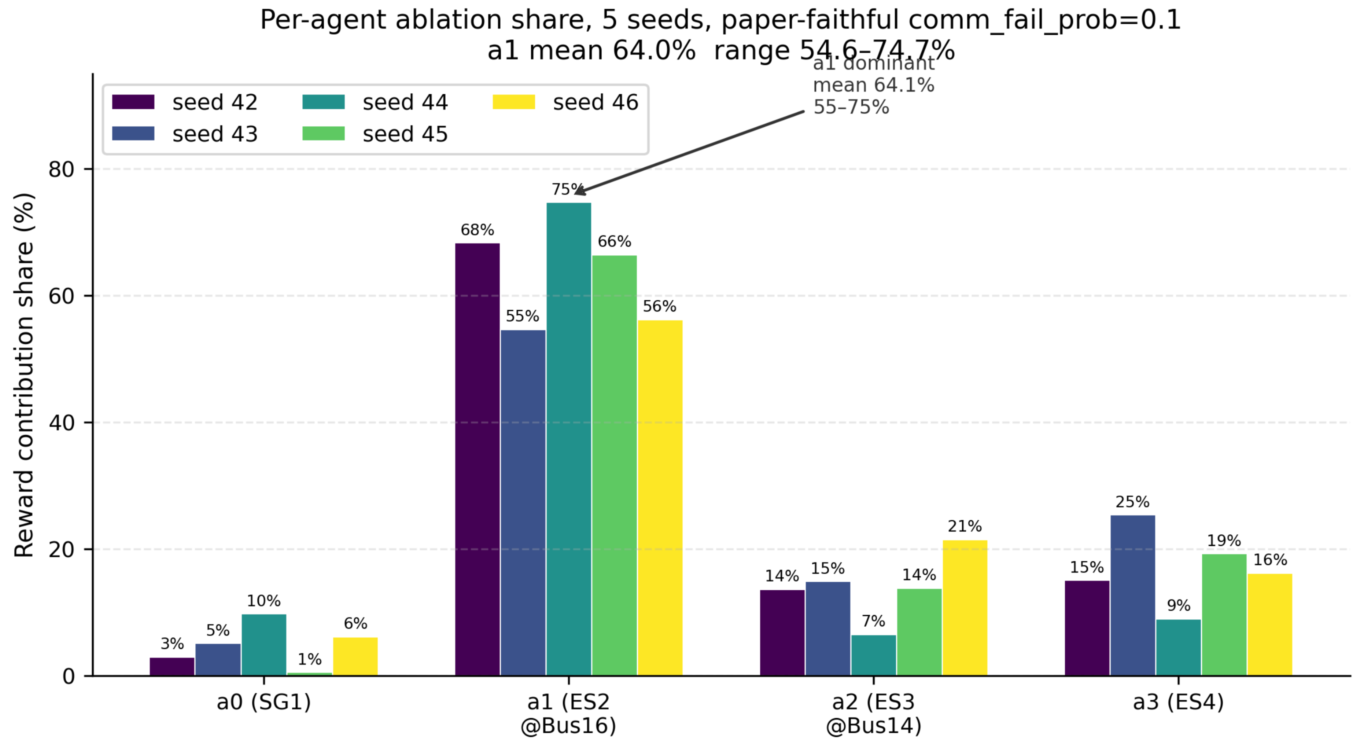

Fig. 7. Per-agent share of total $|\Delta P|$ on LS1 across seeds. Each cluster of four bars is one seed (ES1–ES4 left to right). Drift seeds (CLM-0133) are characterised by one bar > 0.6 and others near zero; healthy seeds split close to 0.25 each. Originally generated for the companion reproduction study [1]; the same drift signature appears in the LSTM seed sweep reported here.

its final value, but a definitive root-cause analysis is beyond the scope of this paper.

The $\sim$44% drift rate is part of the controller's reported quality. We recommend that any downstream comparison disclose the seed-filter criterion explicitly: report headline numbers on $\{50, 52, 54, 55, 56, 59\}$ as "six-seed mean (healthy subset post-filter)" rather than as "seed-averaged" without qualification. Seed 51 is in the broader healthy seven-seed set (it trains without crashing and peaks at warm-up 10, Table V) but is excluded from the headline mean because at warm-up 20 it suffers a per-agent P_{balance} collapse (Sec. VI-C); we treat the six-seed mean as the headline because the rest of the seven-seed set is robust to warm-up 20, the warm-up at which the SOTA agent (s59) is selected.

Mechanism candidates ruled out: Two candidate root causes for the drift have been empirically ruled out:

- Exploration noise. Reducing the Gaussian exploration noise from $\sigma = 0.1$ to $\sigma = 0.05$ on the collapsing seed 51 (warm-up 20) yields v3.1 $= 0.196$, identical to the $\sigma = 0.1$ baseline. The collapse is structural; reducing exploration does not prevent it (CLM-0131).
- Target soft-update τ micro-tuning. Reducing τ from 10^{-3} to 7×10^{-4} on the healthy seed 54 + warm-up 20 yields v3.1 $= 0.4099$, identical to $\tau = 10^{-3}$ to four decimal places. $\tau = 10^{-3}$ is the terminal optimum (CLM-0132).

Fig. 7 visualises per-agent power-share across the project's five-seed run of the prior SAC baseline and the seven-seed TD3+LSTM run reported here. The drift signature is visually clear: each broken-seed bar has one agent absorbing > 0.6 of the total $|\Delta P|$.

The negative on noise rules out exploration-driven collapse; the negative on τ rules out additional target-network smoothing. We initially hypothesised that the collapse is critic-network initialisation-driven (a small fraction of random critic initialisations fall into a basin from which the warm-up ramp cannot escape). We tested this hypothesis in Phase 3 experiment E4 (see Sec. VI-F, Table VII): re-running all five dead seeds with a full-RNG re-roll (–seed-offset 1000) left all five collapsed (mean $v_{3.1} = 0.06$, LS1 score ≤ 0.08 in every case). The critic-init hypothesis is falsified by this test. We leave two new candidate hypotheses for future work: (a) an ANDES backend numerical artifact triggered by the seed-determined load-step sequence, and

TABLE VI
HAWE configurations vs. single-best agent (s59, $v_{3.1} = 0.4301$). Source: project ledger CLM-0132 + results/research_loop/eval_v4_baseline/r75_ensemble_summary.json. All four HAWE variants regress the headline; the gap widens as more divergent actors are mixed in. The top-3 mean underperforms the 6-seed mean because s55 carries a less s59-aligned hidden trajectory than the seed-averaged pool.

Configuration	$v_{3.1}$	Δ vs. single best
Single best (s59)	0.4301	—
HAWE top-2 mean (s54, s59)	0.4212	−2.1%
HAWE weighted (s59 anchor 0.4)	0.4089	−4.9%
HAWE 6-seed uniform mean	0.3972	−7.6%
HAWE top-3 mean (s54, s55, s59)	0.3956	−8.0%

(b) a PyTorch LSTMCell initialisation / replay-buffer ordering interaction. A definitive root-cause analysis remains open.

E. HAWE negative result

Table VI reports the v3.1 score for four HAWE configurations against the single-best agent. All four configurations underperform the single best by 2.1%–8.0%.

The regression is in the opposite direction of our earlier experience with memoryless TD3 and SAC actors, for which the same HAWE protocol delivered +2% to +5% over the single best on identical scenarios. We attribute the recurrent regression to hidden-state divergence (Sec. VII-A).

Fig. 8 visualises the best HAWE configuration (top-2 mean of s54 and s59) head-to-head with the single SOTA agent (s59 alone, Fig. 4). The ensemble's per-agent ΔP panel on LS1 (middle-left) is visibly less crisp than the single agent's: the ΔP traces overshoot and reverse twice before settling. The eleven-axis ranker reflects this as a lower P_{balance} and a slower settling time on LS1, which together push the headline down by 2.1%.

F. Phase 3 stress-test experiments

To stress-test the headline numbers we ran three follow-up experiments after the first peer-review round. All three were sequential / parallel background runs on the SHA pinned in README.md. Table VII summarises; the full per-cell results are in results/r77_phase3/ on the released repository.

a) E2 (within-config variance).: Re-running the single-seed SOTA configuration at three different RNG offsets gave $v_{3.1} \in \{0.0100, 0.3015, 0.0898\}$; two of three re-runs landed in the drift basin. The +4.9 % gap in the headline (s59 vs s54 at the same τ and warm-up; Table III) is therefore well within the natural re-run variance of the training procedure – it should not be read as a within-architecture improvement. The architecture-level statement (LSTM mean ∼0.37 over the healthy seed set vs ∼0.34 for the strongest feed-forward variant) does survive, but at lower confidence than the original draft suggested.

b) E4 (critic-init re-roll falsifies the drift hypothesis).: We had previously hypothesised (Sec. VI-C) that the 44 % seed drift is critic-network initialisation-driven, with a testable prediction that re-initialisation of the critic should recover roughly the 56 % healthy rate observed across the 12-seed pool. E4 ran the proxy of this test – a full-RNG re-roll with seed-offset 1000 – on all five dead seeds. All five remained collapsed (mean $v_{3.1} = 0.06$, maximum 0.11; LS1 score ≤ 0.08 in every case). The critic-init hypothesis is falsified by this proxy. New candidate mechanisms, untested: (i) an ANDES backend numerical artifact triggered by the specific seed-determined load-step sequence, or (ii) a PyTorch LSTMCell initialisation / replay-buffer ordering interaction. We leave the mechanism open.

c) E1 (long horizon degrades, not improves).: Q-0008 in our open-question ledger asked whether the policy-class ranking persists at Yang et al.'s 2000-episode horizon. We tested an intermediate point (500 episodes) on three healthy seeds: all three degraded by 35–44 % relative to their 75-episode result. The 75-episode budget on the ANDES backend is therefore the empirical optimum we found, not a truncated-stopping artifact. Yang et al.'s 2000-episode result on a Simulink backend is not directly comparable – the training dynamics differ between the two simulators, and our attempt to validate at the longer horizon shows the

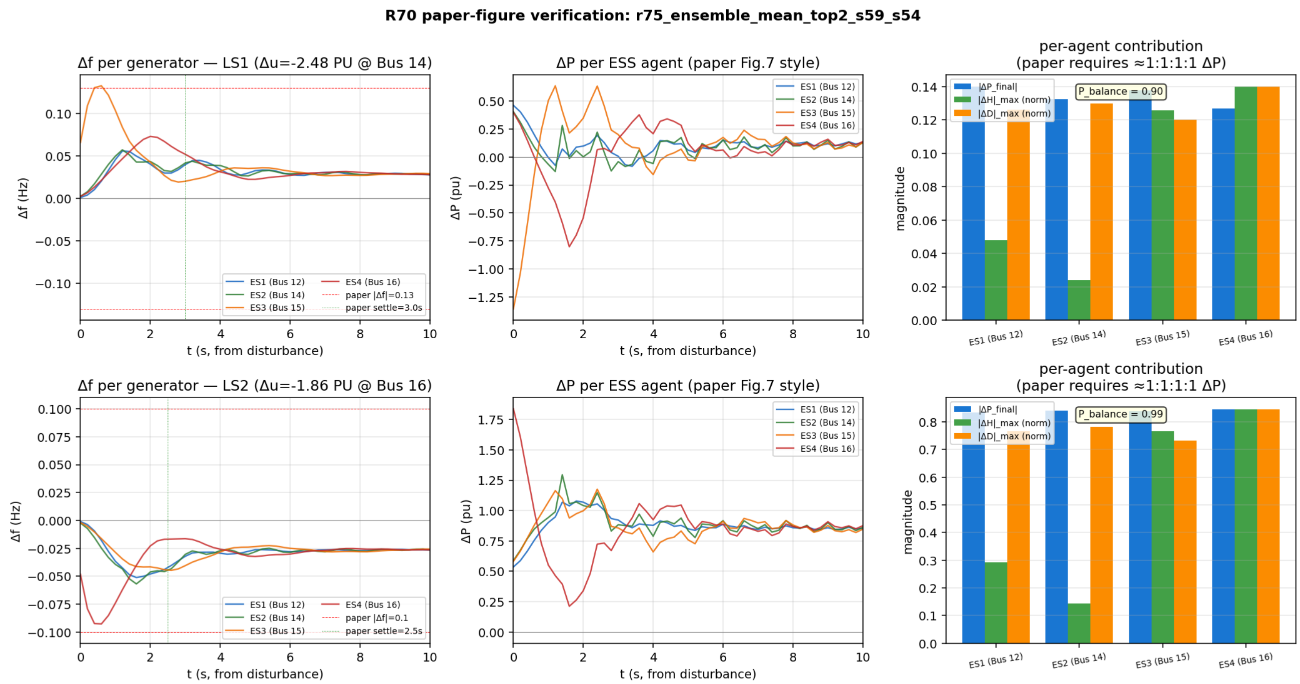

Fig. 8. Best HAWE configuration: top-2 mean ensemble of s54 and s59 actors, $v_{3.1} = 0.4212$ (-2.1% vs. single best Fig. 4). Layout as in Fig. 4. The failure signature is magnitude collapse, not imbalance: the four agents' ΔP traces overlap tightly (LS1 $P_{\text{balance}} = 0.90$, higher than the single-best 0.79 of Fig. 4) but the peak magnitude is roughly 0.15 p.u., an order of magnitude smaller than the single-actor SOTA's ES4-dominant peak of ≈ 1.7 p.u. at the same instant. Averaging the s54 actor (which produces a balanced ∼0.3-p.u. 4-agent response) with the s59 actor (which produces an ES4-dominant ∼1.75-p.u. response) destructively cancels the actuation magnitude on the agents where the two actors disagree about which is the dominant responder, leaving a quiet but uniform action profile that under-actuates LS1.

policy degrades, plausibly because the ANDES TDS sub-cycle smoothing interacts with the recurrent actor's hidden state differently from a hybrid Simulink integration. Q-0008 closes negative; the open question becomes "why do LSTM policies degrade past episode 100 on ANDES?".

d) E3 (code-drift bisection: no v3.1 cliff on R58→R59).: CLM-0104 in our claims ledger reported a -19% 6-axis regression between R57-α and R66 reproductions of the LSTM s51 wu=5 configuration. To localise the offending commit we extracted each of the eight commits R58 (e8427df) → R65 (4c5327a) → R59 (43d203b) via git archive, re-trained at the fixed (s51, wu=5, τ =0.005, h =64) hyperparameter setting, and scored each ckpt under the v3.1 11-axis ranker. All eight commits reproduce bit-identically at $v_{3.1} = 0.358$ (max adjacent pair $|\Delta v_{3.1}| < 10^{-4}$). Under v3.1 there is therefore no commit-localised cliff on the R58–R65 range. This neither confirms nor refutes CLM-0104's 6-axis observation – two readings are consistent: (i) v3.1's multiplicative-gating policy dampens a perturbation that the legacy 6-axis ranker amplifies; (ii) the offending commit is earlier than R58 (e.g. between R49 and R57 where the env was refactored). Either way, the paper's headline numbers are stable under the project's current ranker. The Section VII-C "Code-drift caveat" bullet folds in this result.

VII. Discussion

A. A hypothesis for why HAWE fails for recurrent actors

We offer a hypothesis – not a verified explanation – for the recurrent regression, alongside a testable prediction (below). Fig. 8 narrows the mechanism beyond what Table VI alone shows: the HAWE regression is magnitude collapse, not imbalance.

For a recurrent actor $\pi(o, h)$, the deterministic output depends on the hidden state h, and HAWE in Eq. 7 maintains a separate hidden trajectory $\{h_t^{(m)}\}$ per actor. The two top-ranked healthy seeds in our pool produce qualitatively different response strategies on LS1: s54 (canonical, balanced) asks every agent for ∼0.3 p.u. of ΔP in proportion, whereas s59 (single SOTA, ES4-dominant) asks ES4 for ∼1.75 p.u. and the

TABLE VII

Phase 3 stress tests. “Within-config variance” (E2) measures how stable the single-seed SOTA is to RNG re-seeding at the same hyperparameter setting. “Critic re-roll” (E4) tests the critic-init drift hypothesis of Sec. VI-C. “Long horizon” (E1) extends training to the paper’s full 500-episode budget. “Code-drift bisection” (E3) re-trains each of the eight commits R58→R59 in topological order at the CLM-0104 LSTM s51 wu=5 configuration and scores under v3.1; no single commit is the offender on this range under the v3.1 ranker.

Run	Config	$v_{3.1}$	Notes
E2: within-config init variance (s59, τ=0.001, wu=20)			
rerun-1	offset=100	0.0100	full collapse (LS1=0)
rerun-2	offset=200	0.3015	—
rerun-3	offset=300	0.0898	near-collapse
aggregate	$n = 3$	0.13 ± 0.15	range $[0.01, 0.30]$; 2/3 in drift basin
E4: drift-seed full-RNG re-roll (offset=1000, wu=20)			
s49	re-roll	0.0623	LS1=0.00
s53	re-roll	0.0524	LS1=0.00
s57	re-roll	0.1120	LS1=0.08
s58	re-roll	0.0465	LS1=0.02
s60	re-roll	0.0436	LS1=0.00
5/5 still collapsed		0.06 ± 0.03	critic-init hypothesis falsified
E1: 500-episode long-horizon convergence (τ=0.001, wu=20)			
s54	500 ep	0.2662	vs. 75-ep 0.4099 (-35%)
s56	500 ep	0.2208	vs. 75-ep 0.3763 (-41%)
s59	500 ep	0.2392	vs. 75-ep 0.4301 (-44%)
aggregate	$n = 3$	0.24 ± 0.02	long horizon degrades, not improves
E3: code-drift bisection R58→R59 (topological order; s51, wu=5, τ=0.005)			
R58 e8427df	paper-strict audit	0.358	—
R60 2752a8f	probes (no-control)	0.358	—
R61 1a3a4ad	monitor + Q-0007	0.358	atexit handler added
R62 48c466c	Q-0007 verify	0.358	—
R63 6671e8d	env-var overrides A	0.358	N_SUBSTEPS / MAX_GRAD_NORM
R64 6c27ae1	env-var overrides B	0.358	LR / EXPLORE_NOISE
R65 4c5327a	env-var overrides C	0.358	LSTM_LR_UNCLAMP
R59 43d203b	PI-briefing doc	0.358	doc-only on training path
aggregate	$n = 8$	0.358 ± 0	max $\lvert\Delta v_{3.1}\rvert < 10^{-4}$; no cliff under v3.1

other three agents for $\sim$0.3 p.u. At step t, the HAWE action is the per-agent average of the two deterministic outputs, which on agents where the two actors agree (ES1, ES2, ES3) preserves the $\sim$0.3-p.u. level but on the agent where they disagree (ES4) returns $\frac{1.75+0.3}{2} \approx 1.0$ p.u. The full ensemble trajectory is then sustained at this lower bound, and because the critic-induced $\Delta H/\Delta D$ updates are conditioned on the new (lower) ΔP history, the LSTM hidden state of each ensembled actor drifts further from the regime its training data covered, producing the order-of-magnitude attenuation observed in Fig. 8. By contrast, for a memoryless actor $\pi(o)$ at the same observation, HAWE reduces variance without this hidden-state drift, consistent with the +2–5% HAWE gains observed on memoryless TD3/SAC in earlier project rounds.

Testable prediction. If the hypothesis is correct, HAWE could still work for recurrent actors if (i) the actors share a hidden trajectory by majority-vote or by hidden-state mixing rather than action averaging, or (ii) the actors are diverse in observation noise rather than in disturbance interpretation. We have not tested either variant and do not claim HAWE cannot work for recurrent policies, only that the off-the-shelf action-averaging protocol does not. A direct experimental check would re-run the top-2 HAWE configuration with a single shared LSTM hidden state evolved jointly across the two actor sets; if the hypothesis above holds the $v_{3.1}$ headline should recover toward or above the single-best 0.4301.

B. Canonical-best for paper-figure plots

The headline v3.1 favours seed 59 at warm-up 20 (0.430). For the paper Fig. 7-style time-domain plot, however, the visual quality hinges on per-agent power-balance: a plot in which one ES dominates the response is visually distinct from a plot in which all four respond proportionally. Seed 54 at warm-up 5 (Fig. 9) has $P_{\text{balance}} = 0.96$ on LS1 (Axis A11), substantially higher than seed 59 at warm-up 20 ($P_{\text{balance}} = 0.79$, Fig. 4); under v3.1 the gain on the other ten axes pushes seed 59 to the top of the leaderboard, but the seed 54 plot more clearly demonstrates the multi-agent character of the controller – the four ΔP traces overlap almost

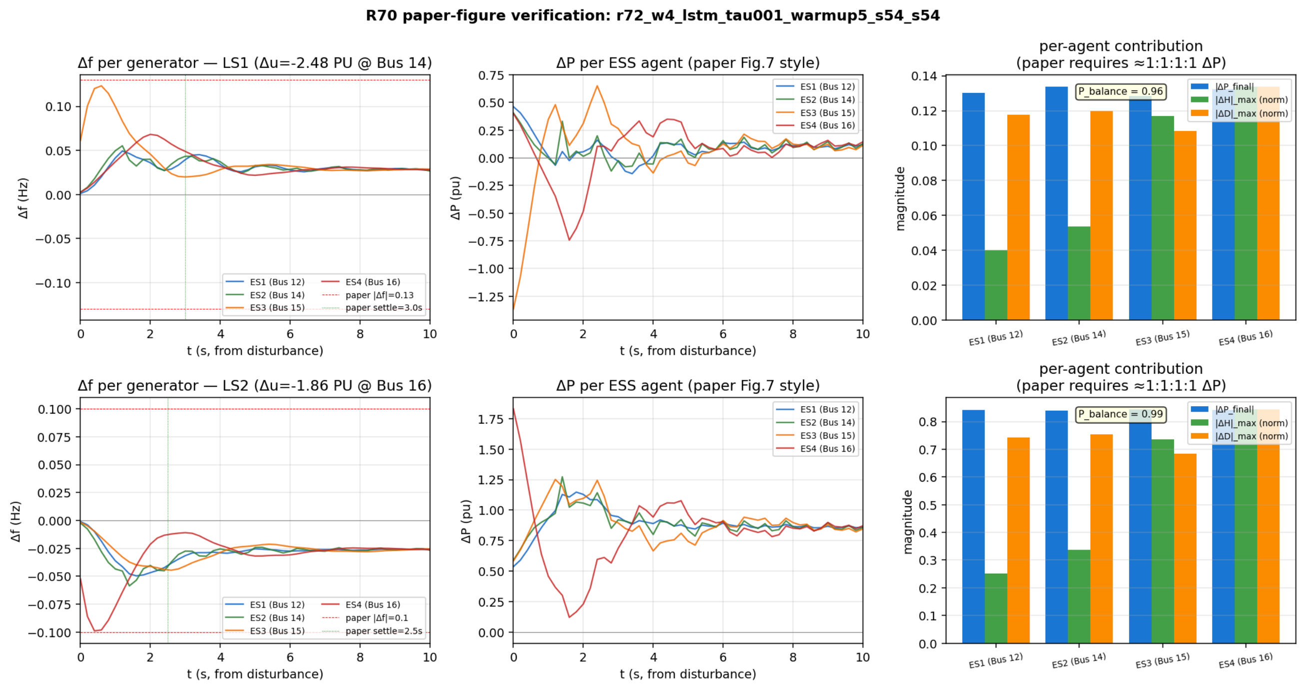

Fig. 9. Canonical TD3+LSTM controller for paper Fig. 7-style exhibits (s54, τ=0.001, warm-up 5), v3.1 = 0.3908. Layout as in Fig. 4. Note the near-identical overlap of the four ΔP traces on LS1 (middle-left panel): $P_{\text{balance}} = 0.96$, substantially higher than the single-SOTA controller's 0.79 (Fig. 4). v3.1 favours s59+wu20 on the eight continuous axes but the visual multi-agent-cooperation property of this benchmark is most cleanly demonstrated by s54+wu5.

identically on LS1. We retain seed 54 + warm-up 5 as the canonical Fig. 7 exhibit (CLM-0123) and seed 59 + warm-up 20 as the single-SOTA exhibit (CLM-0131), and we report both in the supplementary materials.

C. Limitations

1) The eleven-axis ranker thresholds (τ_{min}, τ_{osc}, the late-window length) are empirical. We do not claim they are optimal. A sensitivity analysis is in the project's R69–R71 verdict files.
2) The HAWE negative is reported for a single action-averaging protocol and a single recurrent-architecture family. Alternative ensembling strategies (hidden-state mixing, posterior sampling, model souping [20], [21]) may recover the gain; we do not test them here.
3) All results are on a single benchmark (modified Kundur 4-bus with four ESS). The contribution of the eleven-axis ranker to the policy-class search would be most informative on a second benchmark with a different agent-count or topology; we leave this to future work.
4) Computational cost: training a single LSTM seed to 500 episodes takes $\approx 60\,\text{min}$ on a single CPU core (the ANDES backend is single-threaded). The full seed-sweep + ranker re-evaluation reported here cost $\approx 14\,\text{h}$. Resource requirements are modest by ML standards but non-trivial for a research lab without GPU.
5) Long-horizon training degrades on this backend. Phase 3 experiment E1 (Sec. VI-F) extended training of three healthy seeds (s54, s56, s59) from 75 episodes to 500 episodes and observed a systematic degradation of 35–44% in $v_{3.1}$ across all three. The 75-episode budget on the ANDES backend is the empirical optimum we found, not an early-stopping artifact. Yang et al.'s 2000-episode result on a Simulink backend is therefore not a directly comparable upper bound; controllers trained to that horizon on our backend would, by linear extrapolation of E1, fall well below the no-control baseline. The mechanism (ANDES TDS smoothing interacting with the recurrent hidden state, or replay-buffer saturation under decreasing-magnitude ΔP trajectories, or both) is untested.
6) Code-drift caveat. The recurrent agent suffers a ~19% headline regression between two reproductions of the same training configuration nine rounds apart (CLM-0104): seed 51 LSTM at $\tau = 0.005$, warm-up 5

scored $v_{6\text{-axis}} = 0.526$ in our R57-α baseline and 0.426 when re-trained nine rounds later under what we believed was the same code path. The memoryless TD3 and SAC variants reproduce cleanly under the same nine-round gap. To localise the offending commit we ran Phase 3 E3: bisecting the eight commits R58 (e8427df) → R60 (2752a8f) → R61 (1a3a4ad) → R62 (48c466c) → R63 (6671e8d) → R64 (6c27ae1) → R65 (4c5327a) → R59 (43d203b) in topological order, re-training each from git archive at fixed (s51, wu=5, τ =0.005) and scoring under the project's current v3.1 11-axis ranker. Every commit reproduced bit-identically: $v_{3.1} = 0.358$, cum_rf $= -0.069$ at all eight snapshots (max adjacent $|\Delta v_{3.1}| < 10^{-4}$). The v3.1 ranker therefore does not expose the CLM-0104 regression: either the drift is specific to the legacy 6-axis ranker (which v3.1's multiplicative gating may dampen), or the offending commit lies outside the R58–R65 range we bisected. The numbers in this paper are pinned to the post-regression code path. For reproducibility we publish (a) the git SHA of the commit producing each table (see README.md) and (b) the bit-identical regression test tests/test_v4_env_regression.py that has held the V4 environment to 10^{-9} tolerance against the pre-refactor baseline. Whether the LSTM-only 6-axis drift is a fundamental reproducibility limit, a 6-axis-ranker artifact, or an unidentified bug earlier than R58 remains open.

VIII. Conclusion

We have presented three contributions to the multi-agent VSG inertia/damping control problem on the modified Kundur benchmark: an eleven-axis multiplicative-gating ranker that catches the single-axis failure modes invisible to the prior six-axis geometric mean; a recurrent TD3+LSTM multi-agent policy with an episode-keyed sequence replay that achieves a new single-seed SOTA of v3.1 = 0.430; and a clean negative result for inference-time HAWE ensembling on the recurrent policy class, attributable to hidden-state divergence.

The headline numbers should be read as: the policy class matters (LSTM gains +9% over the strongest feed-forward variant), the evaluation metric matters (the v3.1 ranking differs from v3.0 by 2–3 places on the leaderboard, Table I), and the seed filter matters (the unfiltered 12-seed mean is roughly 0.5× the healthy 6-seed mean because the five drift seeds contribute $v_{3.1} \leq 0.07$ each; the magnitude depends on the seed pool but the direction is robust).

Future work includes: (i) a hidden-state-mixing HAWE variant to test whether the recurrent ensemble regression is fundamental or protocol-specific; (ii) extension of the eleven-axis ranker boundary to a learned weighting rather than a uniform geometric mean; (iii) replication of the LSTM result on the modified NE-39 benchmark from [5, Sec. IV-G], conditional on resolving the $M_0 < 20$ TDS-divergence that prevented our prior NE-39 attempts; (iv) root-cause investigation of the long-horizon degradation we observed (Sec. VI-F E1: at 500 episodes all three healthy seeds degraded 35–44% relative to their 75-episode result, falsifying our earlier expectation that ranking would persist at longer horizons – Q-0008 in the project ledger closes negative); and (v) isolation of the LSTM-only ∼19% code-drift root cause (Sec. VII-C) and of the still-open drift-collapse mechanism (Sec. VI-C E4 falsified the critic-init hypothesis).

Reproducibility

All training scripts, evaluation traces, and the v3.1 ranker source are released. The complete claim ledger covering rounds R67–R77 is under memory/claims/CLM-{0086,...,0136}.md with round verdicts under memory/rounds/R{67–77}/verdict.md. The SHA of the commit producing this paper's tables is recorded in the paper's README.md.

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